light_gbm Â¶

LightGBMConfig Â¶

```
LightGBMConfig(**kwargs)
```

Bases: `GenericLightGBMConfig`

The Microsoft LightGBM configuration object used for model training.

Other Parameters:

Name	Type	Description
`boosting_type`	`str`	Boosting type.
`num_iterations`	`int`	Number of booster iterations.
`learning_rate`	`float`	Learning rate.
`num_leaves`	`int`	Maximum tree leaves.
`lightgbm_seed`	`int`	Seed used to generate other seeds.
`max_depth`	`int`	Maximum tree depth.
`min_data_in_leaf`	`int`	Minimum leaf data points.
`min_sum_hessian_in_leaf`	`float`	Minimal sum of hessian per leaf.
`bagging_fraction`	`float`	Bagging sample fraction.
`bagging_freq`	`int`	Bagging frequency.
`feature_fraction`	`float`	Feature fraction by tree.
`feature_fraction_bynode`	`float`	Feature fraction by tree node.
`lambda_l1`	`float`	L1 regularization.
`lambda_l2`	`float`	L2 regularization.
`min_gain_to_split`	`float`	Minimal gain to split a leaf.

Name	Type	Description
`drop_rate`	`float`	DART drop rate.
`max_drop`	`int`	DART max drop.
`skip_drop`	`float`	DART skip drop.
`xgboost_dart_mode`	`bool`	DART XGBoost mode.
`uniform_drop`	`bool`	DART uniform drop.
`top_rate`	`float`	GOSS top retain rate.
`other_rate`	`float`	GOSS other retain rate.
`min_data_per_group`	`int`	Minimum data per group.
`max_cat_threshold`	`int`	Maximum categories thresholds.
`cat_l2`	`float`	Categorical L2 regularization.
`cat_smooth`	`float`	Categorical smoothing.
`max_cat_to_onehot`	`int`	Max categories to one-hot encode.
`verbosity`	`int`	Verbosity of the LightGBM backend.
`max_bin`	`int`	Max bins.
`min_data_in_bin`	`int`	Minimum bin data.
`bin_construct_sample_cnt`	`int`	Bin construction sample count.
`enable_bundle`	`bool`	Exclusive Feature Bundling (EFB).
`zero_as_missing`	`bool`	Use zeroes as missing.
`is_unbalance`	`bool`	Unbalanced label.
`undersampling`	`float`	The value for undersampling. Only this or oversampling can be used.
`oversampling`	`float`	The value for oversampling. Only this or undersampling can be used.
`seed`	`int`	

Name	Type	Description
		The seed value.
`ignored_fields`	`set of str`	Set of names corresponding to the fields to be ignored when training a model.
`schema`	`Schema`	Pass a schema to enable conversion between descs and internal names. By not passing a schema, ds-api will assume field names are references to internal Pulse names, not the ones in the UI.

Examples:

```
>>> model_config = LightGBMConfig(
...     bagging_fraction=0.9,
...     undersampling=0.85
...)
>>> model_config.num_leaves # gets num_leaves: 31
>>> model_config.num_leaves = 10 # sets num_leaves to 10
>>> model = model_project.train(
...     "model_name",
...     datasource,
...     "target_field",
...     model_config
... )
>>> model.waitUntilModelTrained()
>>> config_from_model = model.get_config()
```

bagging_fraction `property` `writable` Â¶

```
bagging_fraction
```

bagging_freq `property` `writable` Â¶

```
bagging_freq
```

bin_construct_sample_cnt `property` `writable` Â¶

```
bin_construct_sample_cnt
```

boosting_type `property` `writable` Â¶

```
boosting_type
```

cat_l2 `property` `writable` Â¶

```
cat_l2
```

cat_smooth `property` `writable` Â¶

```
cat_smooth
```

drop_rate `property` `writable` Â¶

```
drop_rate
```

enable_bundle `property` `writable` Â¶

```
enable_bundle
```

feature_fraction `property` `writable` Â¶

```
feature_fraction
```

feature_fraction_bynode `property` `writable` Â¶

```
feature_fraction_bynode
```

ignored_fields `property` `writable` Â¶

```
ignored_fields
```

is_unbalance `property` `writable` Â¶

```
is_unbalance
```

lambda_l1 `property` `writable` Â¶

```
lambda_l1
```

lambda_l2 `property` `writable` Â¶

```
lambda_l2
```

learning_rate `property` `writable` Â¶

```
learning_rate
```

lightgbm_seed `property` `writable` Â¶

```
lightgbm_seed
```

max_bin `property` `writable` Â¶

```
max_bin
```

max_cat_threshold `property` `writable` Â¶

```
max_cat_threshold
```

max_cat_to_onehot `property` `writable` Â¶

```
max_cat_to_onehot
```

max_depth `property` `writable` Â¶

```
max_depth
```

max_drop `property` `writable` Â¶

```
max_drop
```

min_data_in_bin `property` `writable` Â¶

```
min_data_in_bin
```

min_data_in_leaf `property` `writable` Â¶

```
min_data_in_leaf
```

min_data_per_group `property` `writable` Â¶

```
min_data_per_group
```

min_gain_to_split `property` `writable` Â¶

```
min_gain_to_split
```

min_sum_hessian_in_leaf `property` `writable` Â¶

```
min_sum_hessian_in_leaf
```

num_iterations `property` `writable` Â¶

```
num_iterations
```

num_leaves `property` `writable` Â¶

```
num_leaves
```

other_rate `property` `writable` Â¶

```
other_rate
```

oversampling `property` `writable` Â¶

```
oversampling
```

seed `property` `writable` Â¶

```
seed
```

skip_drop `property` `writable` Â¶

```
skip_drop
```

subclasses `class-attribute` `instance-attribute` Â¶

```
subclasses: List[JavaWrapper] = []
```

top_rate `property` `writable` Â¶

```
top_rate
```

undersampling `property` `writable` Â¶

```
undersampling
```

uniform_drop `property` `writable` Â¶

```
uniform_drop
```

verbosity `property` `writable` Â¶

```
verbosity
```

xgboost_dart_mode `property` `writable` Â¶

```
xgboost_dart_mode
```

zero_as_missing `property` `writable` Â¶

```
zero_as_missing
```

auto_cast `staticmethod` Â¶

```
auto_cast(config: JavaWrapper) -> LightGBMConfig
```

Creates a LightGBMConfig from the java object.

Parameters:

Name	Type	Description	Default
`config` Â¶	`JavaWrapper`	Java configuration object.	*required*

Returns:

Type	Description
`GeneralizedLinearModelConfig`	Configuration object according to its python representation.